NATIONAL UNIVERSITY OF SINGAPORE

MA3236 - Nonlinear Programming

Final Examination

(Semester 1 : AY2023/2024)

Time allowed : 2 hours

INSTRUCTIONS TO CANDIDATES

1. Please write your matriculation/student number only. Do not write your name.
2. This examination paper contains **5** questions and comprises **4** printed pages.
3. Answer **ALL** questions.
4. Please start each question on a new page.
5. This is a CLOSED BOOK examination.
6. Students are allowed to use one double-sided, A4 size helpsheet.
7. Candidates may use scientific calculators. Graphical and programmable calculators are not allowed. However, they should lay out systematically the various steps in the calculations.

Question 1 [15 marks]

Consider the unconstrained minimization problem:

$$\text{(P)} \quad \min_{\mathbf{x}\in\mathbb{R}^2} f(\mathbf{x}) := \frac{1}{2}\mathbf{x}^T A\mathbf{x} + \mathbf{c}^T\mathbf{x},$$

where $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{c} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.

(a) Show that f is a strictly convex function and determine its unique global minimizer $\mathbf{x}^*$.

(b) Given initial point $\mathbf{x}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, run one iteration of the conjugate gradient method to obtain $\mathbf{x}^{(1)}$ for solving the problem above. Is $\mathbf{x}^{(1)} = \mathbf{x}^*$?

(c) Suppose we reformulate (P) by setting $\mathbf{x} = B\mathbf{y}$ to obtain the following problem:

$$\text{(P')} \quad \min_{\mathbf{y}\in\mathbb{R}^2} \frac{1}{2}\mathbf{y}^T B^T AB\mathbf{y} + \mathbf{c}^T B\mathbf{y}$$

Determine $B \in \mathbb{R}^{2\times 2}$ such that the conjugate gradient method finds the optimal solution of (P') in just one iteration with any initial point, i.e. given any $\mathbf{y}^{(0)} \in \mathbb{R}^2$, $\mathbf{y}^{(1)}$ is the optimal solution of (P'). Justify your answer.

Solution.

(a) $H_f = A$ is positive definite by principle minor test. $\mathbf{x}^* = -A^{-1}\mathbf{c} = -\frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} -1 \\ 0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

(b) $\mathbf{p}^{(0)} = -\nabla f(\mathbf{x}^{(0)}) = -A\mathbf{x}^{(0)} - \mathbf{c} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\alpha_0 = \frac{\|\mathbf{p}^{(0)}\|}{(\mathbf{p}^{(0)})^T A\mathbf{p}^{(0)}} = \frac{1}{2}, \qquad \mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha_0\mathbf{p}^{(0)} = \frac{1}{2}\begin{bmatrix} 1 \\ 0 \end{bmatrix} \neq \mathbf{x}^*$$

(c) Obtaining the minimizer in one step using conjugate gradient method is the same as obtaining the minimizer in one step using steepest descent method. According to the convergence theorem, we need to make $B^T AB = I$.

Here, $A = QDQ^{-1}$, where $D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, $Q = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ So B should satisfy $Q^{-1}B = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{bmatrix}$, implying $B = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$.

We can verify that $B^T AB = I$. $\mathbf{p}^{(0)} = -\mathbf{y}^{(0)} - B^T\mathbf{c}$ and $\mathbf{y}^{(1)} = \mathbf{y}^{(0)} + \alpha_0\mathbf{p}^{(0)} = \mathbf{y}^{(0)} + \alpha_0(-\mathbf{y}^{(0)} - B^T\mathbf{c}) = \mathbf{y}^*$ when $\alpha_0 = 1$.

Question 2 [25 marks]

Suppose $Q \in \mathbb{R}^{n\times n}$ is symmetric positive definite, $A, B \in \mathbb{R}^{m\times n}$ $(m \leqslant n)$ are matrices such that the rows of A and B are linearly independent, $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$. Consider the following problem:

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^n,\, \mathbf{y}\in\mathbb{R}^n} \quad & \tfrac{1}{2}\mathbf{x}^T Q\mathbf{x} + \mathbf{c}^T\mathbf{y} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b}, \\ & B\mathbf{y} = \mathbf{0}. \end{aligned}$$

(a) Give the reason(s) why it suffices to find the KKT points to find the optimal solutions to this problem.

(b) Determine the regularity at every feasible point.

(c) If $(\mathbf{x}^*, \mathbf{y}^*)$ is a KKT point, deduce an explicit formula for $\mathbf{x}^*$ in terms of $A, B, Q, \mathbf{b}$. (You might find it useful that the matrix $AQ^{-1}A^T$ is invertible).

(d) If $(\mathbf{x}^*, \mathbf{y}^*)$ is a KKT point, deduce an explicit formula for $\mathbf{c}^T\mathbf{y}^*$.

(e) Show that the optimal objective function value is $\frac{1}{2}\|\mathbf{b}\|^2$ if $AQ^{-1}A^T = I$ (identity matrix).

Solution.

(a) This problem is a equality constrained convex problem. The global minimizer is KKT point (and vice versa).

(b) We can write the constraint as

$$\begin{bmatrix} A & \mathbf{0} \\ \mathbf{0} & B \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}$$

Here, the rows of $\begin{bmatrix} A & \mathbf{0} \\ \mathbf{0} & B \end{bmatrix}$ are linearly independent since the rows of A and B are linearly independent.

This implies all the feasible points are regular.

(c) KKT system:

$$\begin{cases} Q\mathbf{x} + A^T\lambda = \mathbf{0}, \\ \mathbf{c} + B^T\hat{\lambda} = \mathbf{0} \\ A\mathbf{x} = \mathbf{b} \\ B\mathbf{y} = \mathbf{0} \end{cases} \Longrightarrow \begin{cases} \mathbf{x} = -Q^{-1}A^T\lambda \\ -AQ^{-1}A^T\lambda = \mathbf{b} \end{cases} \Longrightarrow \begin{cases} \mathbf{x} = Q^{-1}A^T(AQ^{-1}A^T)^{-1}\mathbf{b} = \mathbf{x}^* \\ \lambda = -(AQ^{-1}A^T)^{-1}\mathbf{b} \end{cases}$$

(d) From (c), $\mathbf{c} = -B^T\hat{\lambda}$ and $B\mathbf{y}^* = \mathbf{0}$. So $\mathbf{c}^T\mathbf{y}^* = -\hat{\lambda}^T B\mathbf{y}^* = -\hat{\lambda}^T\mathbf{0} = 0$.

(e) From (c), $\mathbf{x}^* = Q^{-1}A^T\mathbf{b}$. From (d), $\mathbf{c}^T\mathbf{y}^* = 0$. Hence, the optimal objective function value is

$$\frac{1}{2}(\mathbf{x}^*)^T Q\mathbf{x}^* + \mathbf{c}^T\mathbf{y}^* = \frac{1}{2}\mathbf{b}^T AQ^{-1}QQ^{-1}A^T\mathbf{b} = \frac{1}{2}\mathbf{b}^T AQ^{-1}A^T\mathbf{b} = \frac{1}{2}\|\mathbf{b}\|^2.$$

Question 3 [25 marks]

Consider the following constrained minimization problem in $\mathbb{R}^n$:

$$\begin{aligned} (\mathrm{P}) \quad \min_{x_1,\, x_2,\, x_3} \quad & x_1 + x_2 + x_3 \\ \text{s.t.} \quad & x_1^2 + x_2 x_3 = 1, \\ & \mathbf{x} \in X := \{\mathbf{x} \in \mathbb{R}^3 \mid x_i \in \{0,1\} \ \ \forall i = 1,2,3\}. \end{aligned}$$

(a) Is this problem convex?

(b) Solve the problem (P) directly.

(c) Let $\lambda \in \mathbb{R}$ be the Lagrange multiplier for the equality constraint. Write down the Lagrangian function and compute the Lagrangian dual function $\theta(\lambda)$.

(d) Solve the Lagrangian dual problem.

(e) Is there any duality gap? Does it contradict with the Strong duality theorem?

Justify your answer in each case.

Solution.

(a) No. X is discrete so it is not a convex problem.

(b) The only feasible points are (1,0,0), (1,0,1), (1,1,0), (0,1,1). Among all these points, (1,0,0) gives the smallest objective function value, with optimal value = 1.

(c) The Lagrangian function:

$$L(\mathbf{x}, \lambda) = x_1 + x_2 + x_3 + \lambda(x_1^2 + x_2 x_3 - 1)$$

For any $\lambda \in \mathbb{R}$, the Lagrangian dual function is:

$$\begin{aligned} \theta(\lambda) &= \inf_{x_1,x_2,x_3 \in \{0,1\}} \{x_1 + x_2 + x_3 + \lambda(x_1^2 + x_2 x_3 - 1)\} \\ &= \min\{-\lambda, 1-\lambda, 2, 1, 3+\lambda\} \\ &= \begin{cases} 3+\lambda, & \text{if } \lambda < -2, \\ 1, & \text{if } -2 \leqslant \lambda < -1, \\ -\lambda, & \text{if } \lambda \geqslant -1 \end{cases} \end{aligned}$$

(d) The Lagragian dual problem is

$$\max_{\lambda \in \mathbb{R}} \{\theta(\lambda)\} = 1$$

attained when $\lambda = -2$.

(e) There is no duality gap since the optimal primal objective function value equal to the optimal dual objective function value, from (b) and (d). It does not contradict with the Strong duality theorem since the problem is not convex.

Question 4 [20 marks]

Consider the following primal problem:

$$\begin{aligned} \text{(P)} \quad \min \quad & f(\mathbf{x}) = -x_1^2 - x_2^2 - x_3^2 \\ \text{s.t.} \quad & x_3 - 2 \leqslant 0 \\ & \mathbf{x} \in \mathbf{X} = \{(x_1, x_2, x_3) : x_1 + x_2 \leqslant 1,\ x_1, x_2 \geqslant 0,\ 0 \leqslant x_3 \leqslant 1\}. \end{aligned}$$

The dual problem is

$$\begin{aligned} \text{(D)} \quad \max \quad & \theta(\mu) \\ \text{s.t.} \quad & \mu \geqslant 0 \end{aligned}$$

where

$$\theta(\mu) = \min\{-x_1^2 - x_2^2 - x_3^2 + \mu(x_3 - 2) : \mathbf{x} \in \mathbf{X}\}.$$

(a) Solve the Lagrangian subproblem when $\mu = 1$ and give the set of minimizers $\mathbf{X}(1)$.

(b) Using (a), compute $\partial\theta(1)$ and find the steepest ascent direction of θ at $\mu^{(0)} = 1$.

(c) Perform the exact line search to find $\mu^{(1)} \geqslant 0$. Is $\mu^{(1)}$ optimal to (D)? Justify your answer.

Solution.

(a) Note that the problem is separable.

$$\theta(\mu) = \min\{-x_1^2 - x_2^2 : x_1 + x_2 \leqslant 1;\ x_1, x_2 \geqslant 0\} + \min\{-x_3^2 + \mu x_3 : x_3 \in [0,1]\} - 2\mu$$

$$\min\{-x_1^2 - x_2^2 : x_1 + x_2 \leqslant 1;\ x_1, x_2 \geqslant 0\} = -1, \quad \text{attained at (1,0) and (0,1).}$$

$$\min\{-x_3^2 + \mu x_3 : x_3 \in [0,1]\} = \min\{0, -1 + \mu\} = 0, \quad \text{attained at 0 and 1 when } \mu = 1.$$

So the set of minimizer $\mathbf{X}(1) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

(b)

$$\partial\theta(1) = conv(\{h(1,0,0), h(0,1,0), h(1,0,1), h(0,1,1)\}) = [-2, -1]$$

The direction of steepest ascent is $d = -1$, which is the subgradient with the smallest Euclidean norm.

(c) $\mu^{(1)} = \mu^{(0)} + \alpha d = 1 - \alpha \geqslant 0$ for a step size $\alpha \geqslant 0$, and

$$\alpha = \arg\max_{\alpha \in [0,1]} \theta(\mu^{(1)}) = \arg\max_{\alpha \in [0,1]} \{-1 + \min\{0, -1 + (1-\alpha)\} - 2(1-\alpha)\} = \arg\max_{\alpha \in [0,1]} \{-3 - \alpha + 2\alpha\} = 1$$

Hence $\mu^{(1)} = 1 - 1 = 0$. $\theta(0) = -1$. Note that $f(1,0,0) = f(0,1,0) = -1$. By weak duality theorem, 0 is an optimal solution of (D), (1,0,0) and (0,1,0) is an optimal solution of (P).

Question 5 [15 marks]

Consider the following constrained optimization problem:

$$\begin{array}{ll} \min & \frac{1}{2}\mathbf{x}^T \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \mathbf{x} + \|\mathbf{x}\|^2 \\ \text{s.t.} & x_1 + x_2 = 1. \end{array}$$

(a) Write down the augmented Lagrangian function $L_A(x_1, x_2, \lambda, \sigma)$, where λ is the Lagrangian multiplier and $\sigma > 0$ is the penalty parameter.

(b) Set $\sigma = 1$. Suppose $x_1^{(0)} = x_2^{(0)} = 0$, $\lambda^{(0)} = 0$. Run one iteration of the augmented Lagrangian method to obtain $x_1^{(1)}, x_2^{(1)}, \lambda^{(1)}$.

(c) Given $A \in \mathbb{R}^{m\times n}$, $\mathbf{b} \in \mathbb{R}^m$, $f : \mathbb{R}^m \to \mathbb{R}$. Consider in general the following constrained optimization problem:

$$\begin{array}{ll} \min\limits_{\mathbf{x}\in\mathbb{R}^n} & f(\mathbf{x}) \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \end{array}$$

Show that the augmented Lagrangian function $L_A(\mathbf{x}, \boldsymbol{\lambda}, \sigma)$ of the problem is strictly convex for any $\sigma > 0$ if f is strictly convex.

Solution.

(a) $L_A(x_1, x_2, \lambda, \sigma) = \frac{1}{2}\mathbf{x}^T \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \mathbf{x} + \|\mathbf{x}\|^2 + \lambda(x_1 + x_2 - 1) + \frac{1}{2\sigma}(x_1 + x_2 - 1)^2$

(b)

$$\begin{aligned} \nabla L_A(\mathbf{x}, \lambda^{(0)}, \sigma) = \mathbf{0} &\Leftrightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2\mathbf{x} + \lambda \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{\sigma}(x_1 + x_2 - 1) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \mathbf{0} \\ &\Leftrightarrow x_1^{(1)} = \frac{2}{5},\ x_2^{(1)} = -\frac{3}{5} \quad \text{when } \sigma = 1 \end{aligned}$$

$$\lambda^{(1)} = \lambda^{(0)} + (x_1^{(1)} + x_2^{(1)} - 1)/\sigma = -\frac{6}{5}$$

(c) The augmented Lagrangian function is

$$L_A(\mathbf{x}, \boldsymbol{\lambda}, \sigma) = f(\mathbf{x}) + \boldsymbol{\lambda}^T(A\mathbf{x} - \mathbf{b}) + \frac{1}{2\sigma}\|A\mathbf{x} - \mathbf{b}\|^2$$

$$\nabla L_A(\mathbf{x}, \boldsymbol{\lambda}, \sigma) = \nabla f(\mathbf{x}) + A^T\boldsymbol{\lambda} + \frac{1}{\sigma}A^T(A\mathbf{x} - \mathbf{b})$$

$$H_{L_A}(\mathbf{x}, \boldsymbol{\lambda}, \sigma) = H_f(\mathbf{x}) + \frac{1}{\sigma}A^TA$$

For any $\mathbf{y} \neq \mathbf{0}$,

$$\mathbf{y}^T H_{L_A}(\mathbf{x}, \boldsymbol{\lambda}, \sigma)\mathbf{y} = \mathbf{y}^T H_f(\mathbf{x})\mathbf{y} + \frac{1}{\sigma}\mathbf{y}^T A^T A\mathbf{y} = \mathbf{y}^T H_f(\mathbf{x})\mathbf{y} + \frac{1}{\sigma}\|A\mathbf{y}\|^2 > 0$$

since H_f is positive definite (because f is strictly convex). Hence, the augmented Lagrangian function is also strictly convex.

End of Paper